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RESEARCH ARTICLE

Integrative Quantitative Proteomics and N-Glycoproteomics Reveal Age-Processing of N-Glycosylation in *Caenorhabditis elegans*

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ABSTRACT

N-glycosylation, a highly diverse post-translational modification, critically regulates organismal aging. The *Caenorhabditis elegans* (*C. elegans*) model is one of the excellent aging models, which exhibits distinctive N-glycosylation features including highly fucosylated core structures, bisecting β1,4-galactose, extensive phosphocholine, and methyl modifications. However, the site-specific and structure-specific landscape of the intact N-glycoproteome in adult nematodes is still unknown. In this study, we generated the first temporal atlas of the N-glycoproteome across the three stages of young adulthood (day 1), mid-life (day 5), and early senescence (day 10) of *C. elegans*, which identified 369 distinct glycoforms at 196 N-glycosites on 161 glycoproteins. A mixed sparse Partial Least Squares Discriminant Analysis (sPLS-DA) model, built from dual-omics data, decoded proteins and N-glycoproteins may act synergistically during the adult age progression. Time-series analysis further elucidated four distinct glycoprotein expression patterns and their functional implications. This represents the largest temporal N-glycosylation dataset of *C. elegans* to our knowledge. A high-throughput identification and quantification pipeline for the intact N-glycopeptides of *C. elegans* has been established. This study highlights key N-glycoproteins and glycans implicated in the regulation of adult age progression and may contribute to the integrative analysis of N-glycoproteome data with other omics datasets.

1 | Introduction

N-glycosylation is one of the most diverse and abundant post-translational modifications [1]. Studies have shown that glycosylation is involved in many biological processes, including growth and development of eukaryotes [2–4]. During the process of growth and development, the glycosylation of individual organisms changes dynamically [5, 6]. *Caenorhabditis elegans* as the first genetically sequenced multicellular organism [7], is an ideal model for the study of aging due to its multi-day life cycle.

There are oligomannosidic, paucimannosidic, truncated complex, fucose-rich, phosphorylcholine-rich (PC-rich), and core chitobiose modifications types of N-glycans in wildtype *C. elegans*, and glycans may be more variable when it comes to mutants [8, 9].

Abbreviations: ABC, ammoniumbicarbonate; AGC, automatic gain control; *C. elegans*, *Caenorhabditis elegans*; DTT, dithiothreitol; ESI, electrospray ionization; FUdR, 5-fluoro-2′deoxyuridine; HCD, higher-energy collisional dissociation; IAA, iodoacetamide; LC-MS, liquid chromatography-mass spectrometry; PC-rich, phosphorylcholine-rich; SDS, sodium dodecyl sulfate; sNCE, stepped normalized collision energy; sPLS-DA, sparse Partial Least Squares Discriminant Analysis; TEAA, triethylammonium acetate buffer; TFA, trifluoroacetic; TMT, tandem mass tag; ZIC-HILIC, zwitterionic hydrophilic interaction liquid chromatography.

Xiaoyu Hu and Weirong Xiang contributed equally.

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Significance of the study

Unraveling the *C. elegans* N-glycoproteome is technically challenging due to its highly heterogeneous and atypical glycan structures, such as extensive core fucosylation and phosphorylcholine substitutions. This study represents a significant methodological advance by establishing a high-throughput identification and quantification pipeline specifically optimized for these intact N-glycopeptides. Beyond the methodology, we present the largest temporal dataset available, serving as a benchmark reference for the community. The application of our dual-omics sPLS-DA model further validates the necessity of integrating glycoproteomics with proteomics to capture the dynamic and complex molecular signatures driving organismal aging.

In contrast to mammals, N-glycans in *C. elegans* present mainly three characteristics. First, core fucosylation is widespread with up to four or five fucose residues found attached to the core chitobiose (GlcNAc-GlcNAc). Additionally, galactosylation can occur on antennae stemming from this core structure [10, 11]. Second, the presence of a bisecting β1,4-linked galactose attached to the core β-mannose is characteristic. Furthermore, this bisecting galactose can sometimes be further modified by a fucose [12]. The third feature involves common post-glycosylation modifications of phosphocholine (PC) modifications and methylations, the former can occur on antennal N-acetylglucosamine, the latter may occur on fucose, mannose, or galactose. Both of core and antennal fucoses can be methylated [13]. Proteomics study of lifespan variation in *C. elegans* revealed that at least one third of proteins turned over more than twice during the aging process [14, 15]. Newly synthesized proteins were also monitored that, with > 70% of these proteins newly synthesized by day 5 of adulthood [16]. Subsequent a study revealed that as the nematode progressed through the L1-L4 stages, the N-glycans gradually became more complex [17]. N-glycan glycoforms of nematodes have also been found by multiple studies subsequently [18]. However, site- and structure-specific changes of N-glycosylation map over adult age progression of *C. elegans* remain to be investigated.

In this study, we carried out site- and structure-specific quantitative N-glycoproteomics characterization of changing of N-glycosylation of *C. elegans* at the ages of day 1, 5, and 10. Up to 797 plausible N-glycan linkages (369 monosaccharide compositions) from 196 N-glycosites of 161 glycoproteins were characterized, which represents the largest temporal N-glycosylation data of *C. elegans*. This analysis revealed dynamic alterations in site-specific N-glycosylation patterns associated with *C. elegans* aging. Quantitative differences in glycopeptide abundance across the three age groups were determined using a TMT labeling strategy [19, 20]. Furthermore, integrating this N-glycoproteomics with proteomics data allowed us to generate preliminary hypotheses and build an exploratory multi-omics model to investigate potential associations with lifespan.

2 | Materials and Methods

2.1 | Reagents and Samples

The 5-fluoro-2′deoxyuridine (FUdR) was purchased from Adamas (Shanghai, China). Agar, peptone, dithiothreitol (DTT), iodoacetamide (IAA), trifluoroacetic acid (TFA, 99%), anhydrous acetonitrile, triethylammonium acetate buffer (TEAA), protease inhibitor cocktail and all HPLC solvents (gradient grade) including acetonitrile (ACN, ≥ 99.9%), methanol (MeOH, ≥ 99.9%) were purchased from Sigma–Aldrich (St. Louis, MO, USA). Bleach was obtained from a local store. TMT 10-plex was obtained from Thermo Fisher Scientific (San Jose, CA, USA). Trypsin (P01001) was purchased from Enzyme & Spectrum (Beijing, China). BCA assay kit, sodium dodecyl sulfate (SDS), ammonium bicarbonate (ABC), and urea were purchased from Sangon Biotech (Shanghai, China). Ultrapure water was produced onsite by a Millipore-QA10 system (Milford, MA, USA).

2.2 | *C. elegans* Culture and Maintenance

Wild-type *C. elegans* (N2) were maintained on Nematode Growth Medium (NGM) plates seeded with *Escherichia coli* OP50 at 20°C. NGM was prepared by dissolving 10.2 g agar, 1.8 g NaCl, and 1.54 g peptone in 600 mL of distilled water, autoclaved, cooled to ~50°C, and supplemented with 1 mM $CaCl_2$, 1 mM $MgSO_4$, 25 mM phosphate buffer, and 5 μg/mL cholesterol. M9 buffer contained 6 g Na_2HPO_4, 3 g KH_2PO_4, and 5 g NaCl per liter, autoclaved, and supplemented with 1 mM $MgSO_4$. *E. coli* OP50 was cultured in LB at 37°C overnight and concentrated ~60-fold by centrifugation.

2.3 | Synchronization and Collection of Worms

Age-synchronized populations were obtained by bleaching gravid adults with a solution of 250 μL bleach, 50 μL 10 M NaOH, and 700 μL ddH_2O for 4 min with gentle agitation. Eggs were collected, washed thoroughly with M9 buffer, and incubated overnight at 20°C to hatch and arrest at the L1 stage. The synchronized L1 larvae were cultured to late L4 stage (defined as day 0 of adulthood), then transferred to NGM plates containing 25 μM FUdR to block progeny. Worms were transferred to fresh plates daily and harvested at day 1, day 5, and day 10 of adulthood for proteomic analysis.

2.4 | Protein Extraction and Tryptic Digestion

Worms were suspended in lysis buffer (4% SDS, 0.1 M Tris/HCl, pH 8.0) and with protease inhibitor cocktail (1% v/v), then treated by ultrasonication. After lysis on ice for 30 min, worms were centrifuged at 14,000 g for 20 min at 4°C, and the supernatant was then mixed with cold acetone (1:5, v/v) at −20°C for 3 h of protein precipitation. Protein concentrations were determined by BCA assay. Proteins (1 mg/ml) were reduced in 10 mM DTT for 20 min at 55°C, alkylated in 20 mM IAA for 30 min at room

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temperature in the dark. Samples were diluted at the ratio of 1:3 with 50 mM ABC and digested with trypsin at a 1:50 enzyme-to-substrate ratio overnight at 37°C. Tryptic peptides were desalted by C18 (Phenomenex Jupiter, 15 μm, 300 Å) solid phase extraction (SPE) columns and dried in a SpeedVac (Thermo Scientific). Peptides were stored at -20°C until further use.

2.5 | TMT-Labeling of Peptides

Desalted peptides from day 1, day 5, and day 10 worms, each with three biological replicates, were quantified using nanodrop (Tiangen). Peptides were dissolved in 0.5 M triethylammonium bicarbonate (TEAB, pH 8.5) and mixed with TMT reagent, which was dissolved in acetonitrile. The nine peptide samples were individually labeled with 126, 127N, 127C, 128N, 128C, 129N, 129C, 130N, and 130C tags. After quenching of the reaction by 5% hydroxylamine, peptides were mixed and desalted using C18 SPE columns and dried in SpeedVac. A small part of the peptides was used directly for proteomics analysis, and the remaining peptides were used for enrichment of intact N-glycopeptides.

2.6 | Enrichment of Intact N-Glycopeptides by Zwitterionic Hydrophilic Interaction Liquid Chromatography (ZIC-HILIC)

The ZIC-HILIC (Merck, 5 μm, 200 Å) SPE column was equilibrated three times with 80% ACN, 1% TFA (v/v). Mixed TMT-labeled peptides were redissolved in 80%ACN, 0.1% TFA (v/v) and loaded on the ZIC-HILIC column five times. At last, intact N-glycopeptides were eluted in 0.1% TFA and 50 mM ABC after washing non-glycopeptides in 80% ACN, 0.1% TFA (v/v). Eluted N-glycopeptides were used for N-glycoproteomics study.

2.7 | C18-RPLC-MS/MS (HCD) Analysis of Intact N-Glycopeptides

A Q-Exactive mass spectrometer (Thermo Fisher Scientific) equipped with an Ultimate 3000 nano-LC system (Thermo Fisher Scientific) was used for LC-MS/MS analysis. TMT-labeled glycopeptides were re-dissolved in mobile phase A (0.2% formic acid in water) and injected into a home-made trap column (200 μm i.d., 5 cm) and separated on an analytical column (75 μm i.d., 65 cm), both columns were packed with C18 resin (Phenomenex Jupiter, 5 μm, 300 Å). The mobile phase flow rate in the analytical column was 300 nL/min; Buffer A was 0.2% formic acid in water, and Buffer B was 0.2% formic acid in 95% acetonitrile. The sample was trapped with 2% Buffer B for 25 min and separated with Buffer B from 2% to 40% for 135 min. Buffer B was then changed to 95% over 10 min, stayed at 95% for 5 min. At last, Buffer B was changed to 2% for equilibration of the analytical column. The eluted peptides were ionized under 2.8 kV, ion transfer capillary temperature was 250°C. The *m/z* range of MS1 was 700–2000 with mass resolution of 70,000. The automatic gain control (AGC) target was 2e5, and the maximum injection time was 50 ms. The Top 20 ions were selected for higher-energy collisional dissociation (HCD) with stepped normalized collision energy (sNCE) of 20%, 30%, 30%. The mass resolution of MS2 was 17,500, and the isolation window was 1.4 m/z. The AGC target and maximum injection time were 5e5 and 250 ms.

2.8 | Identification of Peptides with Fragpipe and Intact N-Glycopeptides with gPeptide

For peptide identification, the LC-MS/MS datasets were searched against the UniProt *C. elegans* proteome database (version 2025-06-26, 4,493 sequences) by Fragpipe-22.0 [21]. 'TMT10' workflow was used. Carbamidomethyl on cystine (C) and TMT label on N-terminal and lysine (K) were considered as fixed modifications. The 'run isobaric quant' option was selected, 'define reference' was set to 'virtual', 'group by' is chosen as 'peptide sequence level', 'normalization' is set to 'MD (median centering)', and the original intensity is not transformed by log2.

For identification of intact N-glycopeptides, the search engine gPeptide (commercialized version of GPSeeker [22]) was adopted; protease was trypsin, up to two missed cleavages were allowed. Carbamidomethyl on cystine (C) and TMT label on N-terminal and lysine (K) were considered as fixed modifications. An enumeration library was constructed based on 186 manually curated *C. elegans* N-glycan core structures (confined to N-X-S/T/C, X≠P). Variable modifications included methylation on hexose/ fucose residues and PC on GlcNAc, generating 19,005 unique glycan structures for database searching. The mass tolerances of the precursor and fragment ions were both 20 ppm. Other parameters were as follows: percentage of matched fragment ions of the peptide backbone ≥ 10%, matched fragment ions of the N-glycan moiety ≥ 1, TopN hits N = 2 (Top1 hits have the lowest p score), Y1 ions Top4. Spectrum-level FDR control 1% was achieved with target-decoy searches. Then, all the IDs were filtered to keep only one intact N-glycopeptide per MS/MS spectrum.

2.9 | Data Process and Visualization

The intensity of the reporter ions for quantitation was normalized to the average intensity of all the reporter ions of the same MS/MS spectrum. The resulting expression ratio matrix was then normalized by the median value of 1 within the sample. The intact N-glycopeptide IDs identified in more than seven out of the nine samples were retained and filled using the random forest method to obtain quantitative IDs. Quantitative data were used for differential expression analysis, subsequent subcellular localization, differential analysis, and cluster analysis.

Given the limited sample size ($n = 9$), we conservatively restricted feature selection to a maximum of three variables per component (keepX = 3) in the single-omics-based sPLS-DA model, as well as the dual-omics-based sPLS-DA model, in order to achieve mutual comparison. All models were implemented in R (version 4.2.3) using the mixOmics (version 6.22.0) with two components optimized via 5-fold cross-validation with 20 repeats.

R version 4.2.3 was used for most of the visualization. The packages and their versions used in building the model are as follows: ggrepel (version 0.9.6), mixOmics (version 6.22.0), ggplot2 (version 3.5.1). Time series analysis is based on the Mfuzz package (version 2.58.0), and hierarchical clustering analysis is based

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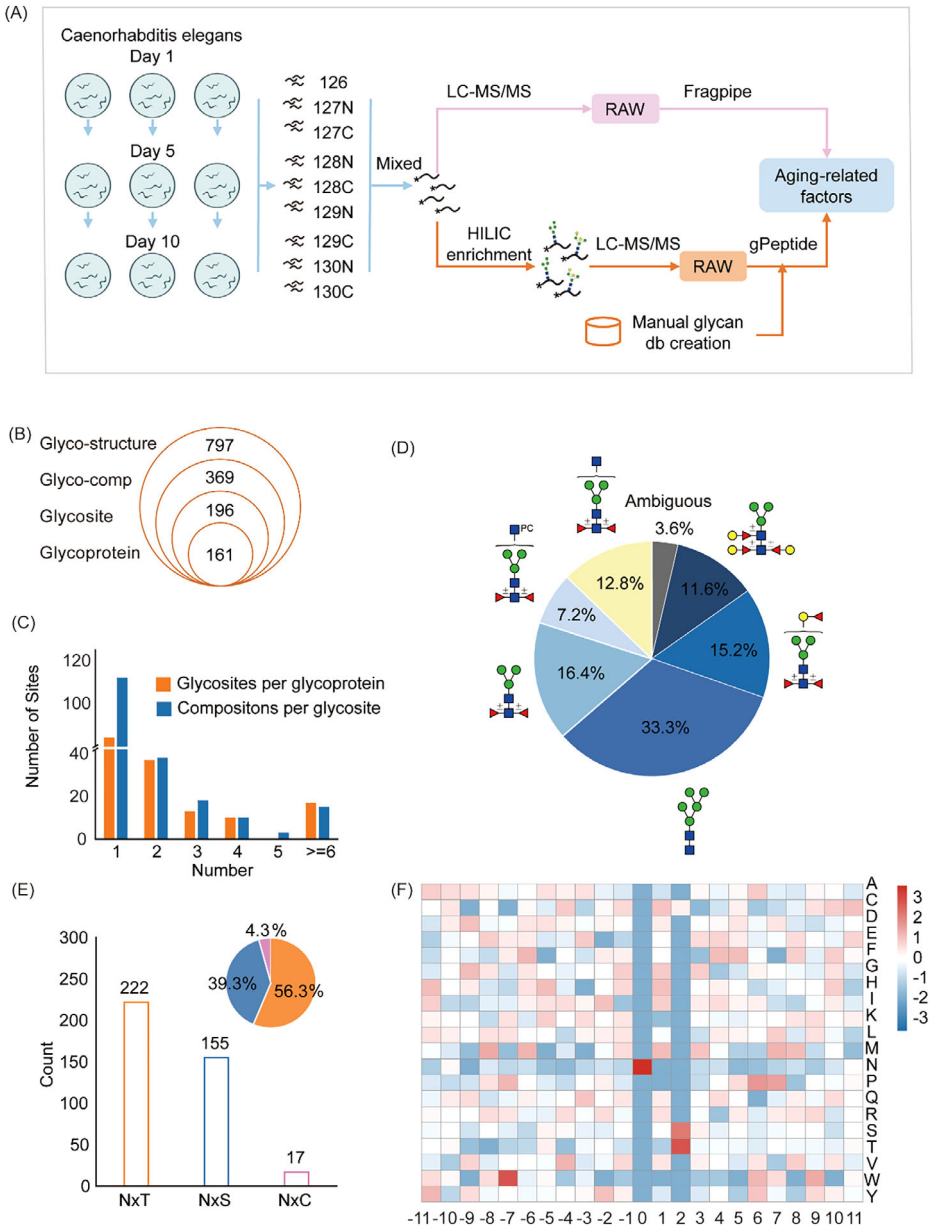


FIGURE 1 | (A) The workflow of this study, three biological replicates were used. (B) The number of identifications at four different levels. (C) The micro- and macro-heterogeneity of N-glycosites. (D) The distribution of the six identified glycan types and the corresponding proportions of their methylation degrees. (E) N-x-S/T/C motif analysis of *C. elegans*. (F) The relative frequency of amino acids adjacent to the N-glycosites (log2 value).

on the pheatmap package (version 1.0.12). The motif analysis was conducted using the online tool WebLogo3, which can be accessed at https://weblogo.threeplusone.com/. Protein-protein interaction analysis with its visualization and bioinformatics analysis were conducted using STRING (https://cn.string-db.org/).

3 | Results

3.1 | Site- and Structure-Specific Profiling of N-Glycosylation in *C. elegans*

With the site- and structure-specific quantitative N-glycoproteomics workflow (Figure 1A), 797 plausible intact N-glycopeptides with 369 monosaccharide compositions were identified from 196 N-glycosites on 161 N-glycoproteins (Figure 1B and Table S1). With the criterium of only one ID per spectrum, 84 N-glycoproteins were identified with one N-glycosite, 37 with two N-glycosites, 13 with three N-glycosites, 10 with four N-glycosites, and 17 with more than six N-glycosites. In terms of identified monosaccharide compositions per N-glycosites, one composition on 112 N-glycosites, two on 38 N-glycosites, three on 18 N-glycosites, four on 10 N-glycosites, and more than six on 15 N-glycosites (Figure 1C). The N-glycans were categorized into six distinct types [23]. High-mannose constituted the most abundant class (33.3%), followed by paucimannose glycans (16.4%). For each group, the glycan types that were identified twice in the three biological replicates were selected, and a relative abundance analysis was conducted, revealing there

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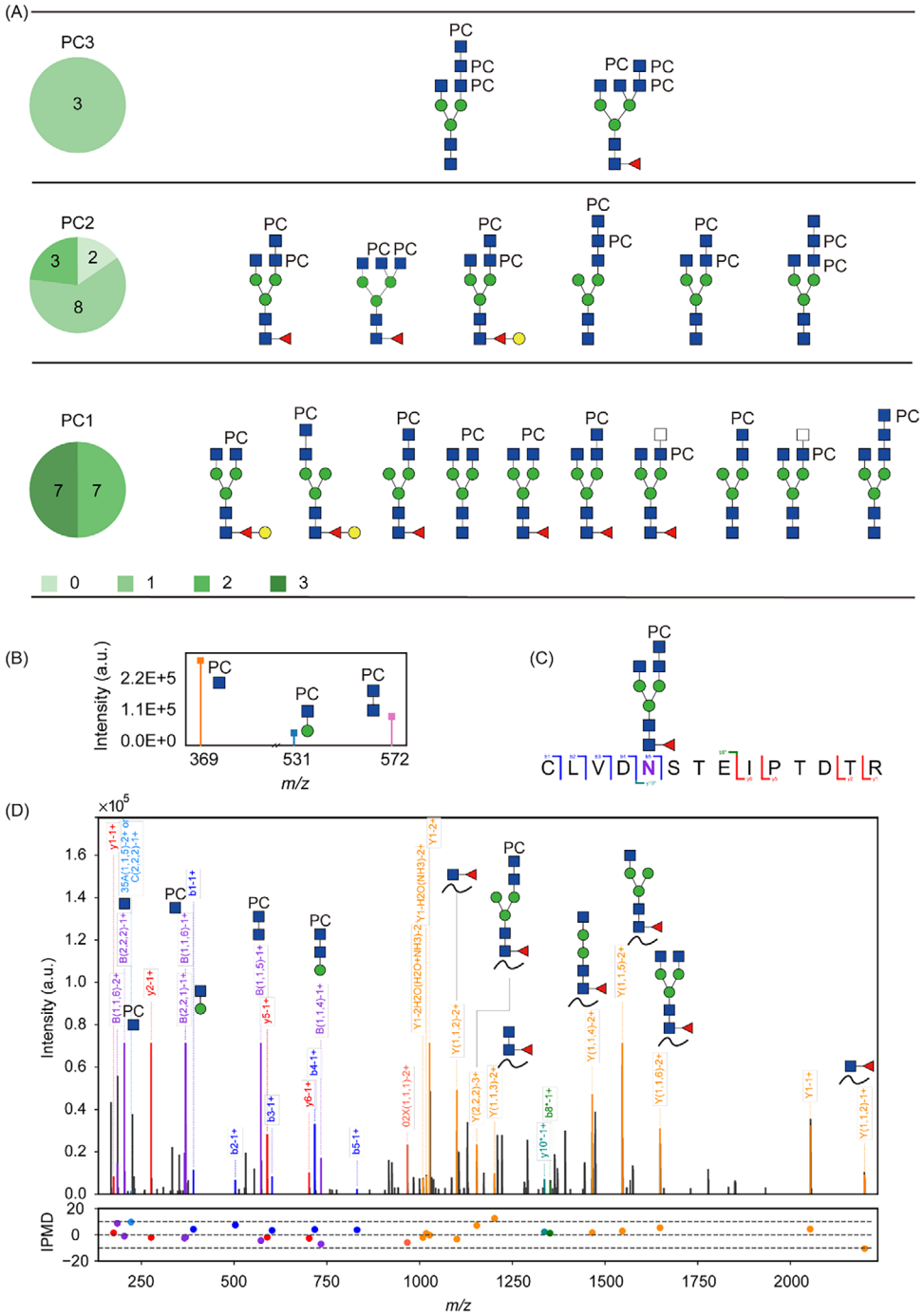


FIGURE 2 | (A) All the linkages of PC-rich glycans and the distribution of the corresponding PC-characteristic ions. Green color intensity represents the number of identified PC-characteristic ions. (B-D) The identification of intact N-glycopeptide CLVDNSTEIPTDTR_5_HexNAc(5)Hex(3)Fuc(1)PC(1) from spectrum 14173; (B) The experimental match of the three PC- feature ions in the MS2 spectrum. (C) The graphical fragmentation map of the peptide backbones with the matched fragment ions and the N-glycan structure; (D) The annotated MS2 spectrum with the matched fragments from both the peptide backbone and N-glycan moiety. Abbreviations: a.u., arbitrary unit; IPMD, isotopic peak m/z deviation.

was a difference of approximately 0.68% in the abundance of Core chitobiose modifications glycans between day 1 and day 5 (**Figure S1**). Motif analysis showed that N-x-T, N-x-S, N-x-C were 56.3%, 39.3%, 4.3% respectively (Figure 1E). Distinct species-specific features were observed as follows: in *C. elegans*, we identified a prominent occurrence of valine (V) at position −4 and significant enrichment of tryptophan (W) at position −7 (Figures 1F, S2A and Table S2). But based on the published data of a large-scale intact N-glycopeptide profiles of mice [24], the occurrence frequency of leucine (L) was relatively high across multiple positions, and there were no significantly enriched residues beyond the N-X-S/T motif (Figure S2B,C). Our findings suggest that the N-glycosylation motif may be less conserved in *C. elegans* compared to vertebrates such as mice.

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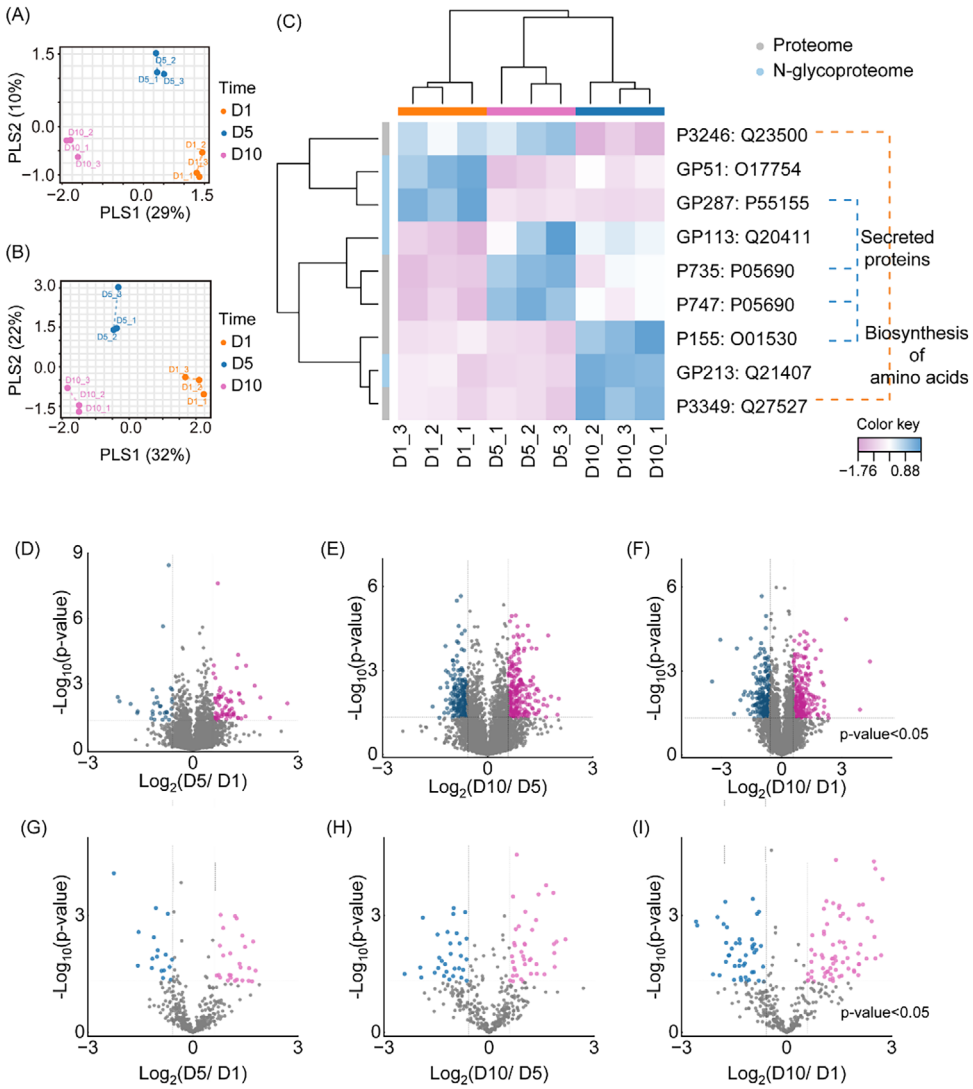


FIGURE 3 | Qualitative classification and quantitative results. (A) sPLS-DA score plot based on the proteome data and (B) the glycoproteome data. (C) Clustered heatmap for the variables selected by mixed sPLS-DA performed on the dual-omics data. (D-F) Volcano plots of the peptides of D5/ D1, D10/ D5, D10/ D1. (G-I) Volcano plots of the N-glycopeptides of D5/D1, D10/D5, D10/D1.

3.2 | Identification of the PC-Rich N-Glycans

PC-rich N-glycans were reported to produce characteristic ions of *m/z* 369 (HexNAc-PC$^+$), 531 (Hex-HexNAc-PC$^+$), and 571 (HexNAc-HexNAc-PC$^+$) [25]. In this study, 30 PC-rich intact N-glycopeptide IDs were identified with 17 plausible linkages (Figure 2A), where a small portion of the core fucose has putative linkages (either α-1,3 or α-1,6), and 29 IDs were identified by the appearance of 1–3 PC-characteristic ions. N-glycans with a maximum of three PC were identified, with the PC-characteristic ion (*m/z* 369) detected in all cases. For intact N-glycopeptide

CLVDNSTEIPTDTR_HexNAc(5)Hex(3)Fuc(1)PC(1), the 369 and 572 PC-characteristic ions closely matched the theoretical values (Figure 2B). Besides, the 734 ion indicated the existence of HexNAc(2)Hex(1)PC(1). By combining a series of Y ions which contained the core fucose, a relatively clear linkage was provided. And the N-glycosite of the glycopeptides was successfully localized by b5* and y10* ions (Figure 2C,D). For another intact N-glycopeptide NNLTLAEALTSFCNVETCN-GAEDNVEK_HexNAc(4)Hex(3)PC(1) characteristic ions at *m/z* 369 and 531 were both detected (Figure S3A), the glycosite was unambiguously assigned using eight site-determining fragment ions (such as b3*, Figure S3B,C).

3.3 | Coordinated Molecular Signatures in Aging and Differential Expression of Peptides and N-Glycopeptides

Distinct sparse partial least squares discriminant analysis (sPLS-DA) models were trained based on the proteomic and N-glycoproteomic data (Table S3), respectively. The N-glycoproteomics model achieved superior group separation with two latent components compared to the proteomics model. The first latent component of the N-glycoproteomics model explained 32% of the variance in group discrimination,

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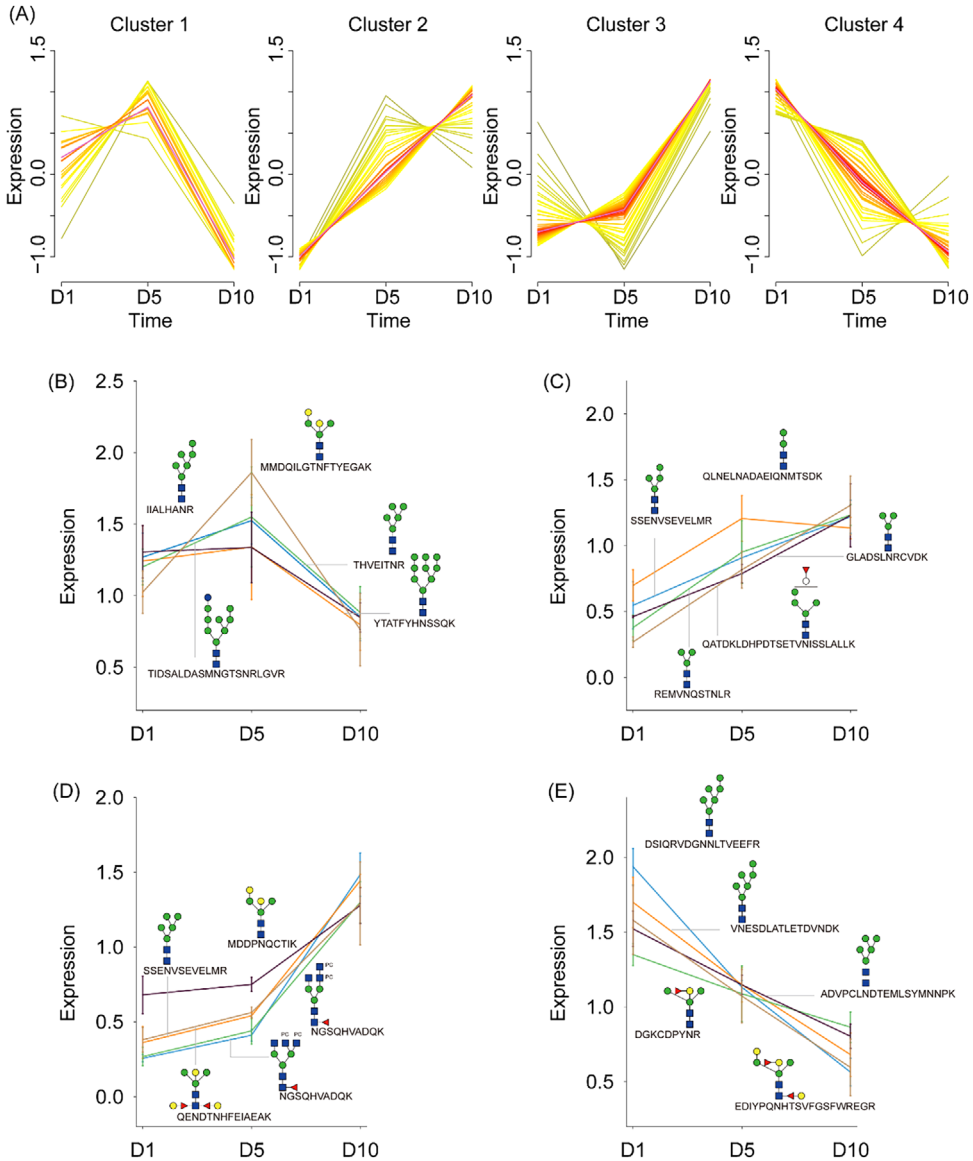


FIGURE 4 | (A) Temporal analysis of differentially expressed intact N-glycopeptides. (B-E) Representative structures and expression fold changes of key intact N-glycopeptides from clusters 1-4.

while the second component explained an additional 22%; the corresponding indicators for the proteomics model were 29% and 10%, respectively (Figure 3A,B). To investigate potential synergistic effects between the two omics, an integrated dual-omics sPLS-DA model was subsequently trained. Mean misclassification error rates analysis showed that the error rate was below 0.2, demonstrating better performance than individual omics models (Figure S4). Despite this sample constraint, the clustering heatmap revealed pronounced inter-group differentiation (Figure 3C). Significantly, all dual-omic clusters identified by mixOmics co-contained unmodified peptides and intact N-glycopeptides, demonstrating systematic co-regulation between proteins and N-glycoproteins. Further biological validation was achieved by projecting the highly loaded features onto the STRING database. Significantly enriched functional terms (**Table S4**) included extracellularly secreted proteins (FDR = 0.0126) and biosynthesis of amino acids (FDR = 0.0467). Secreted proteins require glycosylation for proper folding and trafficking [26, 27], explaining co-variation between peptide abundance and glycan modifications. Collectively, dual-omics sLPS-DA model captured biologically coherent signatures. The identified protein classes may serve as critical nodes (**Figure S5**) for cross-omics regulation in the adult age progression of *C. elegans*.

Corresponding to the coordinated molecular signatures captured by the sPLS-DA models, both the proteome and the N-glycoproteome of *C. elegans* underwent significant changes over aging under the standard laboratory culture conditions, with uniform FUdR treatment applied across all age groups to suppress progeny and ensure synchronization. 757 peptides from 302 proteins exhibited significant alterations (Figures 3D–F and Table S5), representing over 30% of the detected proteins, which is consistent with a previous study [14]. Besides, peptides from gene *rpl-9* showed downregulated in day 10 versus day 5, consistent with the conclusion of previous study, which showed

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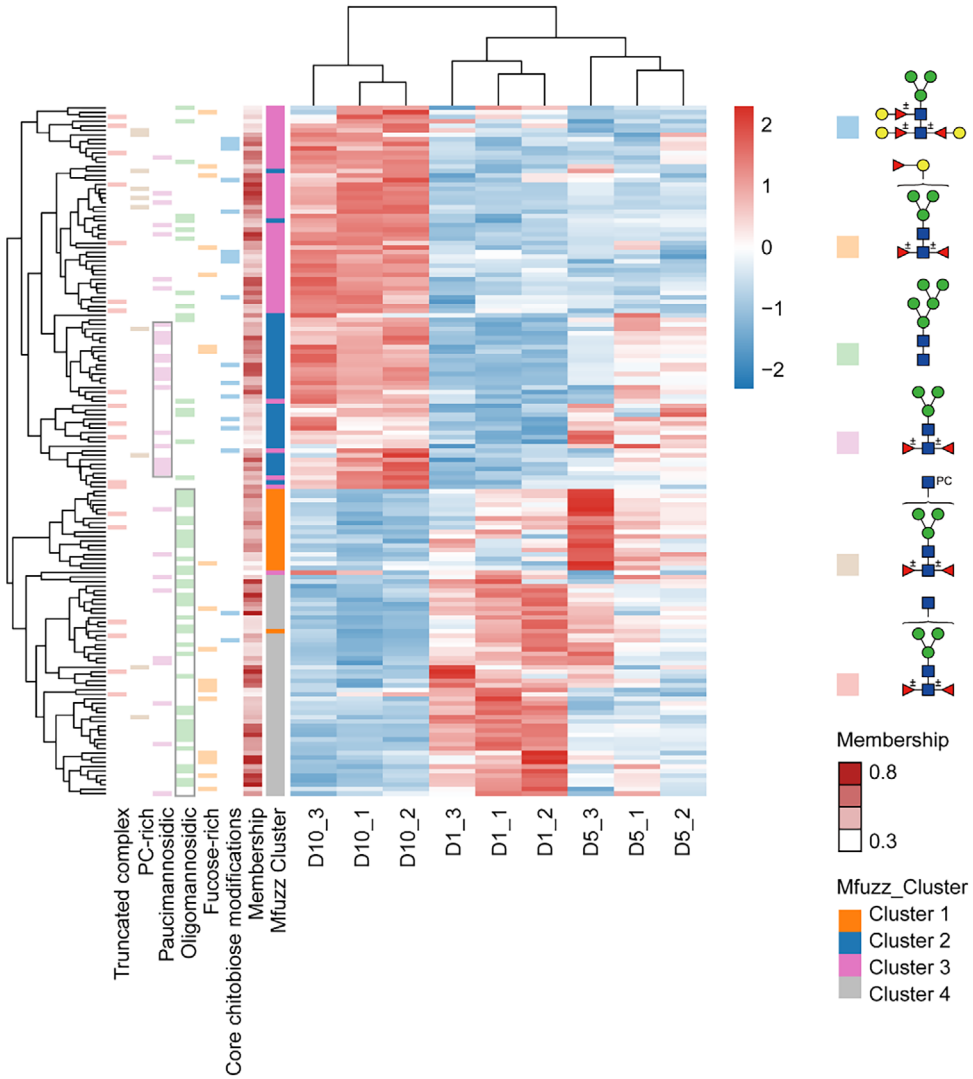


FIGURE 5 | The heatmap combined with the clustering results of the temporal analysis of all differentially expressed intact N-glycopeptides.

C. elegans life extended after knocking down *rpl-9* [16]. There were 153 intact N-glycopeptides derived from 89 N-glycoproteins that exhibited significant alterations (Figure 3G–I and Table S6). More than 50% N-glycoproteins were significantly changed, indicating substantial remodeling and reconstruction.

3.4 | Temporal Analysis of Differentially Expressed Intact N-Glycopeptides and Visualization

According to the temporal changes, the differential intact N-glycopeptides were grouped into four clusters (Figure S6). Heterogeneous N-glycosylation showed four trends (Figure 4A and Table S7). Cluster 1 had an inverted U-shape, Cluster 2 was continuously increasing, Cluster 3 had delayed acceleration, and Cluster 4 was continuously decreasing. Representative top-ranked intact N-glycopeptide structures from each cluster were displayed in Figure 4B–E. Bisecting galactose plays a functionally significant role across Clusters 1, 3, and 4. Here, some of the gal-fuc are classified as the bisect type. When HexNAc(1)Hex(1)Fuc(1) fragment is present, it will be located in the core part (Figure S7–9). In addition, galactoseβ1-4fucose structures, unique in invertebrates [28], were observed in all the four clusters, suggesting regulatory importance in aging-related pathways.

Hierarchical clustering was adopted for the validation of the Mfuzz classification and simultaneous characterization of the distribution profile of all the differentially expressed intact N-glycopeptides (Figure 5). The heatmap showed that the clustering results were basically consistent with the Mfuzz classification. The predominant glycoforms in both Cluster 1 and Cluster 4 were the oligomannosidic type. In contrast, Cluster 2 displayed a relative enrichment of paucimannosidic glycans. The glycoproteins from each cluster underwent comprehensive functional enrichment analysis using the STRING database (Table S7).

Cluster 1, associated with nutrient storage and transport (Figure S10), showed transient upregulation of glycoproteins involved in nutrient reservoir activity (GO:0045735) and lipid transporter activity (GO:0005319), peaking at day 5 before declining to day 10. Notably, vit, a highly glycosylated lipid transporter, and related proteins (Vitellogenin-2, Vitellogenin-4, Vitellogenin-6) exhibited strong interactions (Figure S11A). Consistent with a prior report that vit-2 inhibition is linked to longevity in daf-2

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mutants and vit knockdown extends lifespan in wild-type worms [29], our data found an early upregulation phase preceding age-related decline. This points to candidate pathways that may influence aging phenotypes, although direct evidence on lifespan was not obtained in this study.

Cluster 2 was majorly enriched in embryonic developmental and metabolic lethality (Figure S10). Embryo development ending in birth or egg hatching (GO:0009792) and regulation of cell cycle (GO: 0051726) were enriched in biological process. Glycoproteins from genes egg-1, vit-2, and clec-91 show strong interactions (Figure S11B). Notably, egg-1, which was required for *C. elegans* fertility [30], assembled with the CBD-1 complex to regulate yolk layer hardening [31].

Cluster 3 glycoproteins functioned in cellular morphogenesis and structural integrity (Figure S10). Cell adhesion (GO:0007155), cell morphology variant (WBPhenotype:0000533), and alimentary system development variant (WBPhenotype:0000617) were enriched, which were highly correlated with various variations within the cells. Protein-protein interactions showed integrin cell surface interactions, featuring disorganized muscle protein 1 (DIM-1) (Figure S11C). Notably, DIM-1 knockout extends *C. elegans* lifespan [32] though phenotypic divergence emerges only after day 10 in long-term studies (0–30 days). Our study showed DIM-1 expression dynamics from day 1 to 10 preceding this delayed phenotype, which may provide a new perspective.

Cluster 4 was majorly enriched as locomotion regulation and behavioral plasticity. Many positive regulation processes, such as positive regulation of multicellular organismal process (GO:0051240) and positive regulation of locomotion (GO:0040017), were enriched (Figure S10). In the protein-protein interactions in Cluster 4 (Figure S11D), we identified the microsomal triglyceride transfer protein G5ECG7, which is encoded by dsc-4. Knockdown of this gene is known to result in excessive intestinal fat accumulation [33, 34].

4 | Discussion

This study delivered the first comprehensive temporal atlas of site-specific N-glycosylation dynamics during adult age progression in wild-type *C. elegans* under laboratory culture conditions. By implementing an integrated TMT-9plex workflow across nine samples spanning three lifespan stages (days 1, 5, and 10), we achieved comprehensive characterization of the wild-type N-glycoproteome. High-resolution LC-MS/MS enabled precise mapping of phosphocholine modifications, resolving part of the structural features while quantifying 369 distinct glycoforms across 196 glycosites on 161 glycoproteins. Integrated proteomic and glycoproteomic profiling revealed coordinated molecular remodeling during aging. Through differential expression analysis combined with sPLS-DA modeling and hierarchical clustering, we observed stage-specific regulatory networks that identify key biomolecules. Our analysis identified four glycoprotein clusters with distinct functional associations. Their expression patterns further suggest potential regulation by genes known to be involved in lifespan regulation. Although widely used for synchronized adult sampling, FUdR may influence certain aging-related phenotypes. Besides, this study is a discovery-based investigation aimed at generating hypotheses regarding N-glycosylation in adult age progression, and further orthogonal validation is required to confirm the regulatory factors identified here [35–37]. Collectively, this work still provided foundational insights into how N-glycosylation dynamics contribute to the aging process and delivers a multi-omics resource for further investigation of post-translational regulation in senescence.

Author Contributions

Z.T conceived the study and revised the manuscript. W.X. and X.Z. prepared *C. elegans*, and X.H. did all the other experiments and wrote the draft. X.H. and S.Q. did data analysis.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The raw LC-MS/MS datasets from analysis of intact N-glycopeptides enriched from the *C. elegans* of three different ages have been uploaded to the ProteomeXchange (https://www.proteomexchange.org/) with the identifier number PXD067150.

References

1. A. Varki, "Biological Roles of Oligosaccharides: All of the Theories Are Correct," *Glycobiology* 3 (1993): 97–130.

2. T. Takeuchi, Y. Arata, and K.-I. Kasai, "Galactoseβ1-4fucose: A Unique Disaccharide Unit Found in N-Glycans of Invertebrates Including Nematodes," *Proteomics* 16 (2016): 3137–3147.

3. X. L. Ding, D. Y. Shang, Y. Cui, et al., "Tandem HILIC-IMAC Strategy for Simultaneous N-Glycoproteomics and Phosphoproteomics in Aging Mouse Brain," *Journal of Chromatography A* 1739 (2025): 11.

4. J. C. Sha, J. T. Fan, R. R. Zhang, et al., "B-Cell-Specific Ablation of β-1,4-Galactosyltransferase 1 Prevents Aging-Related IgG Glycans Changes and Improves Aging Phenotype in Mice," *Journal of Proteomics* 268 (2022): 10.

5. R. S. Haltiwanger and J. B. Lowe, "Role of Glycosylation Indevelopment," *Annual Review of Biochemistry* 73 (2004): 491–537.

6. H. T. Feng, S. Y. Lim, A. K. C. Laserna, et al., "High Throughput Human Plasma N-Glycan Analysis Using DNA Analyzer and Multivariate Analysis for Biomarker Discovery," *Analytica Chimica Acta* 995 (2017): 106–113.

7. C. elegans Sequencing Consortium., "Genome Sequence of the Nematode *C. elegans*: A Platform for Investigating Biology," *Science* 282 (1998): 2012–2018.

8. K. Paschinger, S. Yan, and I. B. H. Wilson, "N-Glycomic Complexity in Anatomica Simplicity: *Caenorhabditis elegans* as a Non-Model Nematode?" *Frontiers in Molecular Biosciences* 6 (2019): 9.

16159861, 2026, 9, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/pmic.70133 by Harvard University, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

9. E. Jankowska, L. M. Parsons, X. Z. Song, D. F. Smith, R. D. Cummings, and J. F. Cipollo, "A Comprehensive *Caenorhabditis elegans* N-Glycan Shotgun Array," *Glycobiology* 28 (2018): 223–232.

10. A. J. Hanneman, J. C. Rosa, D. Ashline, and V. N. Reinhold, "Isomer and Glycomer Complexities of Core GlcNAcs in *Caenorhabditis elegans*," *Glycobiology* 16 (2006): 874–890.

11. W. B. Struwe and V. N. Reinhold, "The Conserved Oligomeric Golgi Complex Is Required for Fucosylation of N-Glycans in *Caenorhabditis elegans*," *Glycobiology* 22 (2012): 863–875.

12. S. Yan, L. Brecker, C. S. Jin, et al., "Bisecting Galactose as a Feature of N-Glycans of Wild-Type and Mutant *Caenorhabditis elegans*," *Molecular & Cellular Proteomics* 14 (2015): 2111–2125.

13. K. Paschinger and I. B. H. Wilson, "Comparisons of N-Glycans Across Invertebrate Phyla," *Parasitology* 146 (2019): 1733–1742.

14. D. M. Walther, P. Kasturi, M. Zheng, et al., "Widespread Proteome Remodeling and Aggregation in Aging *C. elegans*," *Cell* 161 (2015): 919–932.

15. T. Xia, E. R. Horton, A. E. Salcini, R. Pocock, T. R. Cox, and J. T. Erler, "Proteomic Characterization of *Caenorhabditis elegans* Larval Development," *Proteomics* 18 (2018): 7.

16. K. Vukoti, X. K. Yu, Q. H. Sheng, et al., "Monitoring Newly Synthesized Proteins Over the Adult Life Span of *Caenorhabditis elegans*," *Journal of Proteome Research* 14 (2015): 1483–1494.

17. I. B. H. Wilson, S. Yan, C. Jin, et al., "Increasing Complexity of the N-Glycome During Caenorhabditis Development," *Molecular & Cellular Proteomics* 22 (2023): 100505.

18. B. Eckmair, C. Gao, A. Y. Mehta, et al., "Recognition of Highly Branched N-Glycans of the Porcine Whipworm by the Immune System," *Molecular & Cellular Proteomics* 23 (2024): 19.

19. H. L. Yang and Z. X. Tian, "Sialic Acid Linkage-Specific Quantitative N-Glycoproteomics Using Selective Alkylamidation and Multiplex TMT-Labeling," *Analytica Chimica Acta* 1230 (2022): 13.

20. A. D. Xue, L. L. Kong, J. L. Li, et al., "Comprehensive Host Cell Proteins Profiling in Biopharmaceuticals by a Sensitivity Enhanced Mass Spectrometry Strategy Using TMT-Labeling and Signal Boosting," *Analytica Chimica Acta* 1335 (2025): 9.

21. A. T. Kong, F. V. Leprevost, D. M. Avtonomov, D. Mellacheruvu, and A. I. Nesvizhskii, "MSFragger: Ultrafast and Comprehensive Peptide Identification in Mass Spectrometry-Based Proteomics," *Nature Methods* 14 (2017): 513.

22. K. Xiao and Z. Tian, "Site- and Structure-Specific Quantitative N-Glycoproteomics Using RPLC-pentaHILIC Separation and the Intact N-Glycopeptide Search Engine GPSeeker," *Current Protocols in Protein Science* 97 (2019): 94.

23. K. Paschinger, M. Gutternigg, D. Rendic, and I. B. H. Wilson, "The N-Glycosylation Pattern of *Caenorhabditis elegans*," *Carbohydrate Research* 343 (2008): 2041–2049.

24. P. Fang, X. M. Yu, M. Y. Ding, et al., "Ultradeep N-Glycoproteome Atlas of Mouse Reveals Spatiotemporal Signatures of Brain Aging and Neurodegenerative Diseases," *Nature Communications* 16 (2025): 16.

25. K. Paschinger, D. Rendic, G. N. Lochnit, V. Jantsch, and I. B. H. Wilson, "Molecular Basis of Anti-Horseradish Peroxidase Staining in *Caenorhabditis elegans*," *Journal of Biological Chemistry* 279 (2004): 49588–49598.

26. Y. Kim, M. S. Lee, H. D. Kim, and J. Kim, "Ribosomal Protein S3 (rpS3) Secreted From Various Cancer Cells Is N-Linked Glycosylated," *Oncotarget* 7 (2016): 80350–80362.

27. B. Zhou, F. Liu, Z. D. Zhu, et al., "N-Glycosylation Is Required for Efficient Secretion of a Novel Human Secreted Glycoprotein, hPAP21," *FEBS Letters* 576 (2004): 401–407.

28. T. Takeuchi, M. Tamura, K. Ishiwata, et al., "Galectin-2 Suppresses Nematode Development by Binding to the Invertebrate-Specific Galactoseβ1-4fucose Glyco-Epitope," *Glycobiology* 29 (2019): 504–512.

29. C. T. Murphy, S. A. McCarroll, C. I. Bargmann, et al., "Genes That Act Downstream of DAF-16 to Influence the Lifespan of *Caenorhabditis elegans*," *Nature* 424 (2003): 277–284.

30. W. L. Johnston and J. W. Dennis, "The Eggshell in the *C. elegans* Oocyte-to-Embryo Transition," *Genesis* 50 (2012): 333–349.

31. D. P. González, H. V. Lamb, D. Partida, et al., "CBD-1 Organizes Two Independent Complexes Required for Eggshell Vitelline Layer Formation and Egg Activation in *C. elegans*," *Developmental Biology* 442 (2018): 288–300.

32. M. H. Rothi, G. C. Sarkar, J. A. Haddad, et al., "The 18S rRNA Methyltransferase DIMT-1 Regulates Lifespan in the Germline Later in Life," *Nature Communications* 16 (2025): 6944.

33. J. Zhang, S. Hashmi, F. Cheema, et al., "Regulation of Lipoprotein Assembly, Secretion and Fatty Acid β-Oxidation by Kruppel-Like Transcription Factor, *Klf-3*," *Journal of Molecular Biology* 425 (2013): 2641–2655.

34. A. S. DePina, W. B. Iser, S.-S. Park, S. Maudsley, M. A. Wilson, and C. A. Wolkow, "Regulation of *Caenorhabditis elegans* Vitellogenesis by DAF-2/IIS Through Separable Transcriptional and Posttranscriptional Mechanisms," *BMC physiology* 11 (2011): 11.

35. J. Hirabayashi, K. Hayama, H. Kaji, T. Isobe, and K. Kasai, "Affinity Capturing and Gene Assignment of Soluble Glycoproteins Produced by the Nematode *Caenorhabditis elegans*," *Journal of Biochemistry* 132 (2002): 103–114.

36. X. L. Fan, Y. M. She, R. D. Bagshaw, J. W. Callahan, H. Schachter, and D. J. Mahuran, "Identification of the Hydrophobic Glycoproteins of *Caenorhabditis elegans*," *Glycobiology* 15 (2005): 952–964.

37. D. F. Zielinska, F. Gnad, K. Schropp, J. R. Wisniewski, and M. Mann, "Mapping N-Glycosylation Sites Across Seven Evolutionarily Distant Species Reveals a Divergent Substrate Proteome Despite a Common Core Machinery," *Molecular Cell* 46 (2012): 542–548.

Supporting Information

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